Name_____ Date_____ Period_____

**LESSON 6-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2**

# Writing Repeated Factors as Powers

Maryland Standard 6.EE.A.1 · Learning target – I can write repeated multiplication as a power and name its base and exponent.

**REMEMBER**

- A **power** shows a repeated factor: $5 \times 5 \times 5 = 5^3$.
- The **base** is the repeated factor. The **exponent** tells how many times it is a factor.
- Read $5^3$ as "5 to the third power" or "5 cubed." Read $5^2$ as "5 squared."

**WORKED EXAMPLE**

Write $6 \times 6 \times 6 \times 6 \times 6$ as a power. Name the base and the exponent.

**Step 1** The repeated factor is 6. It is the base.

**Step 2** Count the factors: there are 5. That is the exponent.

**Step 3** Write the base with a raised exponent: $6^5$

**Answer** **$6^5$; base 6, exponent 5**

**WORD BANK**

**power** – a short way to write a repeated factor, like $4^3$

**base** – the number that repeats

**exponent** – the small raised number: how many times the base repeats

## PRACTICE

*show your work · label units*

**1.** Write each product as a power.

**a.** $4 \times 4 \times 4 =$ _____

**b.** $9 \times 9 =$ _____

**c.** $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$ _____

**d.** $10 \times 10 \times 10 \times 10 =$ _____

**e.** $1/3 \times 1/3 \times 1/3 =$ _____

**f.** $0.5 \times 0.5 =$ _____

**g.** $6 \times 6 \times 6 \times 6 \times 6 \times 6 =$ _____

**h.** $2/5 \times 2/5 =$ _____

**2.** Complete the table.

**Step 1** Base = the repeated number.

**Step 2** Exponent = how many times it is a factor.

| Power | Base | Exponent | Expanded form |
|---|---|---|---|
| **$8^3$** | | | |
| | 7 | 4 | |
| | | | 11 × 11 |

**3.** **Phone Tree.** A coach calls **3** players. Each player calls **3** more, and so on. In round 5, the number of calls is **3 × 3 × 3 × 3 × 3**. Write this as a **power**.

**Step 1** The repeated factor is the base.

**Step 2** Count the 3s for the exponent.

**Power**_____

**4.** **Find the Error.** Rafael wrote 4 × 4 × 4 × 4 × 4 as $5^4$. Explain his mistake. Then write the correct power.

**Frame** *Rafael switched the ___ and the ___. The correct power is ___.*

_____

_____

_____

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** Explain to someone at home why $2^4$ does not mean 2 × 4.

Name_____ Date_____ Period_____

**LESSON 6-3 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4**

# Finding the Value of a Power

Maryland Standard 6.EE.A.1 · Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows.

**REMEMBER**

- Write the **expanded form**, then multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, **not** $4 \times 3 = 12$.
- Fraction and decimal bases work the same way: $(1/2)^2 = 1/2 \times 1/2 = 1/4$.

**WORKED EXAMPLE**

Find the value of $2^5$.

**Step 1** Expanded form: $2 \times 2 \times 2 \times 2 \times 2$

**Step 2** Multiply: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$, $16 \times 2 = 32$

**Answer** **$2^5 = 32$**

**WORD BANK**

**evaluate** – find the value

**expanded form** – the power written as multiplication: $4^3 = 4 \times 4 \times 4$

**squared** – to the 2nd power

**cubed** – to the 3rd power

## PRACTICE

*show your work · label units*

**1.** Find the value of each power.

**Step 1** Write each power in expanded form.

**Step 2** Multiply one factor at a time.

**a.** $7^2$ = _____ **b.** $3^4$ = _____ **c.** $5^3$ = _____

**d.** $10^5$ = _____ **e.** $2^6$ = _____ **f.** $1^8$ = _____

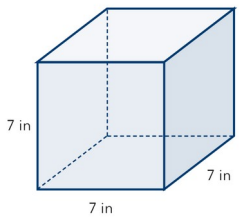
**2.** Find the value. Write fractions in simplest form.

**a.** $(1/2)^3$ = _____ **b.** $0.3^2$ = _____

**3.** Write each number as a power of the given base.

**a.** 27 as a power of 3: _____ **b.** 10,000 as a power of 10: _____

**4. Storage Cube.** A storage cube has edges that are **7 inches** long. Its **volume** is $7^3$ cubic inches. Find the volume.




**Step 1** Write $7^3$ as 7 × 7 × 7.

**Step 2** Multiply 7 × 7 first, then multiply by 7.

*Show your work.*

**Answer**_____ $in^3$

**5. Compare.** Select the **two** powers that have the same value.

**Step 1** Find the value of each power.

**Step 2** Pick the two with the same value.

☐ **A** $2^6$ ☐ **B** $6^2$ ☐ **C** $4^3$ ☐ **D** $3^4$

## TEST PRACTICE

*choose the best answer*

**6.** What is the **value** of $4^3$? Write 4 × 4 × 4 first.

**A** 7 **B** 12 **C** 64 **D** 81

**7.** Which expression has a value of 1,000?

**A** $10^3$ **B** $3^{10}$ **C** 10 × 3 **D** $100^3$

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** Ask someone at home to guess which is greater, $2^5$ or $5^2$. Then find both values together.